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IV-BCSAD

Scenario:

You're working on a personal project named "my_local_project" and want to experiment with new features without affecting the main codebase.

Note:

- Ensure Git is installed on your machine.

```
C:\Users\Anne>git --version
git version 2.46.0.windows.1
```

- Set up your Git username and email (if not already configured)

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project (master)
$ git config --global user.name "Polaris"

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project (master)
$ git config --global user.email "polaris.09a@gmail.com"
```

- Create a directory called my_local_project, navigate into it, and initialize it as a Git repository

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests
$ mkdir my_local_project

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests
$ cd my_local_project

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project
$ git init
Initialized empty Git repository in C:/Users/Anne/Desktop/GitTests/my_local_proj
ect/.git/
```

- Multiple commits reflecting changes in different branches.
- Evidence of conflict resolution during branch merging (optional).
- A Git tag-marking version v1.0 (optional).

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests
$ mkdir my_local_project

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests
$ cd my_local_project

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project
$ git init
Initialized empty Git repository in C:/Users/Anne/Desktop/GitTests/my_local_project/.git/

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project (master)
$ git add .

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project (master)
$ git config --global user.name "Polaris"

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests/my_local_project (master)
$ git config --global user.email "polaris.09a@gmail.com"

```

Tasks:

1. Create a new branch:

- Create a new branch named "**feature_x**" to isolate your experimental changes.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (master)
$ git branch feature_1

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (master)
$ git branch feature_2

```

2. Stage changes:

- Make modifications to your project files.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_1)
$ git status
On branch feature_1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   my_local_project/index.html

```

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_2)
$ git status
On branch feature_2
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   my_local_project/index.html

```

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_3)
$ git status
On branch feature_3
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   my_local_project/index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_4)
$ git status
On branch feature_4
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   my_local_project/index.html
```

- Stage the changes using the git add command.

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_1)
$ git add .

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_1)
$ git status
On branch feature_1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   my_local_project/index.html
```

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_2)
$ git add .

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_2)
$ git status
On branch feature_2
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   my_local_project/index.html
```

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_3)
$ git add .

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_3)
$ git status
On branch feature_3
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   my_local_project/index.html
```

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_4)
$ git add .

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_4)
$ git status
On branch feature_4
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   my_local_project/index.html

```

3. Commit changes:

- Commit the staged changes to your local repository using the git commit command. Provide a clear and concise commit message.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_1)
$ git commit -m "Uncommented a line in index.html"
[feature_1 95f179b] Uncommented a line in index.html
1 file changed, 1 insertion(+), 1 deletion(-)

```

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_2)
$ git commit -m "Uncomment the line h1 that says hi in index.html"
[feature_2 2f728c5] Uncomment the line h1 that says hi in index.html
1 file changed, 1 insertion(+), 1 deletion(-)

```

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_3)
$ git commit -m "3rd brachn commit?"
> 
[feature_3 2a82eba] 3rd brachn commit?
1 file changed, 2 insertions(+)

```

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_4)
$ git commit -m "4th branch commit"
[feature_4 ce0b7c0] 4th branch commit
1 file changed, 4 insertions(+)

```

4. Switch back to the main branch:

- Use the git checkout command to switch back to the main branch.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_2)
$ git checkout master
Switched to branch 'master'

```

5. Merge changes from the feature branch:

- Use the git merge command to merge the changes from the "feature_x" branch into the main branch. Resolve any merge conflicts that may arise.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (master)
$ git merge feature_1
Updating 09a2c09..95f179b
Fast-forward
 my_local_project/index.html | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

```

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (master)
$ git merge feature_2
Auto-merging my_local_project/index.html
Merge made by the 'ort' strategy.
 my_local_project/index.html | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

6. Viewing the Commit History:

- Use Git commands to view the commit history and ensure that all commits are in place.

```
Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (feature_4)
$ git log
commit ce0b7c0349b28aa93a28617ade545cea70c9ce95 (HEAD -> feature_4)
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 15:07:56 2024 +0800

    4th branch commit

commit e0c2f1e1b12062105abfbfb0d70439473ddcbbba (master)
Merge: 95f179b 2f728c5
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 15:00:05 2024 +0800

    Merge branch 'feature_2'

    This is my second merge.

commit 2f728c5004a3b1d72ddffb2b83f605695e4bb679 (feature_2)
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 14:54:19 2024 +0800

    Uncomment the line h1 that says hi in index.html

commit 95f179b406cb989d7c0fe9b446bfabc0a58ad8e7 (tag: v1.0, feature_1)
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 14:50:46 2024 +0800
:
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 15:07:56 2024 +0800

    4th branch commit

commit e0c2f1e1b12062105abfbfb0d70439473ddcbbba (master)
Merge: 95f179b 2f728c5
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 15:00:05 2024 +0800

    Merge branch 'feature_2'

    This is my second merge.

commit 2f728c5004a3b1d72ddffb2b83f605695e4bb679 (feature_2)
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 14:54:19 2024 +0800

    Uncomment the line h1 that says hi in index.html

commit 95f179b406cb989d7c0fe9b446bfabc0a58ad8e7 (tag: v1.0, feature_1)
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 14:50:46 2024 +0800

    Uncommented a line in index.html

commit 09a2c092969c84d798b12a18d589a973c7af5385
Author: Polaris <polaris.09a@gmail.com>
Date: Mon Sep 23 14:45:30 2024 +0800

    First File
```

```
Author: Polaris <polaris.09a1@gmail.com>  
Date: Mon Sep 23 15:07:56 2024 +0800
```

4th branch commit

```
commit e0c2f1e1b12062105abfbfb0d70439473ddcbbba (master)  
Merge: 95f179b 2f728c5  
Author: Polaris <polaris.09a1@gmail.com>  
Date: Mon Sep 23 15:00:05 2024 +0800
```

Merge branch 'feature_2'

This is my second merge.

```
commit 2f728c5004a3b1d72ddffb2b83f605695e4bb679 (feature_2)  
Author: Polaris <polaris.09a1@gmail.com>  
Date: Mon Sep 23 14:54:19 2024 +0800
```

Uncomment the line h1 that says hi in index.html

```
commit 95f179b406cb989d7c0fe9b446bfabc0a58ad8e7 (tag: v1.0, feature_1)  
Author: Polaris <polaris.09a1@gmail.com>  
Date: Mon Sep 23 14:50:46 2024 +0800
```

Uncommented a line in index.html

```
commit 09a2c092969c84d798b12a18d589a973c7af5385  
Author: Polaris <polaris.09a1@gmail.com>  
Date: Mon Sep 23 14:45:30 2024 +0800
```

First File

~

```

(END)
Author: Polaris <polaris.09a1@gmail.com>
Date: Mon Sep 23 15:07:56 2024 +0800

    4th branch commit

commit e0c2f1e1b12062105abfbfb0d70439473ddcbbba (master)
Merge: 95f179b 2f728c5
Author: Polaris <polaris.09a1@gmail.com>
Date: Mon Sep 23 15:00:05 2024 +0800

    Merge branch 'feature_2'

    This is my second merge.

commit 2f728c5004a3b1d72ddffb2b83f605695e4bb679 (feature_2)
Author: Polaris <polaris.09a1@gmail.com>
Date: Mon Sep 23 14:54:19 2024 +0800

    Uncomment the line h1 that says hi in index.html

commit 95f179b406cb989d7c0fe9b446bfabc0a58ad8e7 (tag: v1.0, feature_1)
Author: Polaris <polaris.09a1@gmail.com>
Date: Mon Sep 23 14:50:46 2024 +0800

    Uncommented a line in index.html

commit 09a2c092969c84d798b12a18d589a973c7af5385
Author: Polaris <polaris.09a1@gmail.com>
Date: Mon Sep 23 14:45:30 2024 +0800

    First File

```

7. Tagging Versions (Optional):

- After completing the merge, use the git tag command to create a tag named v1.0 to mark the first version of your project.

```

Anne@LENOVO-LOQ MINGW64 ~/desktop/GitTests (master)
$ git tag v1.0

```